Alejandro Sánchez Yalí

AI Full Stack Engineer

Let's Connect

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SUMMARY

AI Full Stack Engineer with 8+ years designing and shipping web applications end to end and 3+ years building LLM-powered products. Architect and lead engineer of three deployed systems – an AI WhatsApp support platform, an AI-assisted recruitment platform and a surgical supply chain system – spanning Python/FastAPI backends, Next.js/TypeScript frontends, PostgreSQL and cloud deployment. M.Sc. in Mathematics and 11 years teaching mathematics and physics, which shows up in design docs and code review.

TECHNICAL SKILLS

- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL
- **Frontend:** React.js (7+ years), Next.js (6+ years), Tailwind CSS, Shadcn UI, Redux, Zustand, Zod, Effect, Three.js
- **Backend:** FastAPI, Django, Django REST Framework, Celery, NestJS, Node.js, Express, RESTful APIs, GraphQL, microservices
- **AI & LLMs:** LangChain, LangGraph, LiteLLM, RAG, MCP, OpenAI API, Whisper, ElevenLabs, PyTorch, TensorFlow
- **Databases & Infra:** PostgreSQL, Redis, MongoDB, Firebase, Docker, AWS, GCP, Terraform, GitHub Actions, CI/CD
- **Engineering:** Clean architecture, DDD, TDD, Vitest, Jest, React Testing Library, code review

EXPERIENCE

Independent AI Consultant Oct 2026 – Present
Freelance *Remote*

– Lead engineer of Aluna, an AI recruitment platform for staffing agencies (see below)

AI Specialist Aug 2025 – Sep 2026
Lapzo *Remote, Mexico*

– Designed and built a Digital Professor that holds real-time conversations with learners and explains course content through interactive slides
– Owned the architecture connecting LLM services, ElevenLabs voice synthesis and the real-time layer built with Next.js, NestJS and Firebase Realtime Database
– Built an HR agent (LangGraph, FastAPI, PostgreSQL) that answers policy questions with citations and executes HR requests with human confirmation
– Built the AI content-generation pipelines (LangChain, LangGraph, n8n), with Celery workers running generation in the background

Independent Software & AI Consultant Aug 2024 – Jul 2025
Freelance *Remote*

– Lead engineer of CREARIA's multi-tenant AI support agent on WhatsApp, email and Instagram, in production with several client businesses including a real-estate agency: led a team of three across 5 FastAPI services and 2 Next.js dashboards on AWS ECS with Terraform, RAG over pgvector, 11 MCP tools scoped per tenant, LiteLLM fallback routing and human escalation; load-tested to 250 virtual users. Also designed the architecture of an AI coaching agent for one of CREARIA's clients
– Delivered end-to-end web applications for business clients: React.js/Next.js/TypeScript on the front, Python/FastAPI/Django and Node.js/NestJS behind it, deployed on AWS with CI/CD and Stripe payments
– Ran technical design sessions with clients and made the technology decisions for each project; code reviewer for a client's development team

Full Stack Engineer Apr 2021 – Aug 2024
Monadical *Remote, Canada/US*

– Full-stack engineer on ZoHuddle, a video conferencing app for scientific conferences (Neuromatch) that matches participants by mutual interests and supports conferences of about 200 participants: React front end and Django back end, with Django Channels (WebSockets) and Twilio Video

– Full-stack engineer on Virtue Poker, a decentralized poker platform on Ethereum supporting up to 50 concurrent tables of 8 players: React front end, Django back end and smart-contract integration
– Built VC Tools, an LLM tool that retrieves and processes company information for investors (OpenAI, LangChain)
– Reviewed code on a distributed team across Canada, the US and Latin America

Full Stack Engineer Aug 2018 – Oct 2020
BCFort *Medellín, Colombia*

– Built blockchain dApp frontends (React, Web3.js) and analytics platforms (Django, Node.js) on AWS, GCP and IBM Cloud

Professor of Mathematics & Physics (part-time) Jan 2010 – Oct 2021
Universidad de Antioquia *Medellín, Colombia*

– Taught mathematics and physics; in the last year and a half also machine learning and data science with Python, NumPy, Pandas, TensorFlow and PyTorch
– Developed the AI/ML elective curriculum for the Mathematics Institute and mentored student research projects

FEATURED PROJECTS

- **Plixiq – AI Customer Support on WhatsApp** – *Python, FastAPI, LiteLLM, PostgreSQL, Redis, Next.js, WebSocket*
  – Architect and lead engineer of a multi-tenant SaaS where AI agents answer routine customer inquiries, book appointments and escalate the rest to human operators with full conversation context
  – Modular monolith (14 components plus a shared kernel) with boundaries enforced in CI; LiteLLM layer with a primary and fallback provider per agent, so an outage becomes a fallback
- **VitaStock – Surgical Supply Chain for Clinics** – *Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway*
  – Architect and lead engineer of a platform commissioned by a private surgical clinic in Colombia – 12 modules and 5 user roles designed to replace spreadsheet-based inventory and procurement; delivered, with rollout at the clinic pending
  – Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex
- **Aluna – AI Recruitment Platform for Staffing Agencies** – *Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API*
  – Lead engineer of an AI recruitment platform for staffing agencies – 336 source files across 4 packages, 26 tables and 11 external providers – split into a Next.js 15 application owning all domain state and a stateless FastAPI service owning model calls and the WhatsApp channel
  – Durable Inngest analysis pipeline (9 checkpointed steps, 3 retries) where deduplication runs before quota enforcement so a re-analysed CV reuses its cached score instead of paying for another model call; one screening engine serves WhatsApp, web chat and email

OPEN SOURCE

- **talking-avatar-with-ai** – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus** – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion

TALKS & WRITING

- **Scaling Learning Models: Parallelism Strategies in JAX with Flax** – PyCon Colombia, 2024
- **Engineering case studies:** Plixiq · CREARIA Agent · Aluna – architecture and trade-offs behind each system (2026)

EDUCATION

M.Sc. in Mathematics Aug 2010 - Sep 2013
Universidad de Antioquia *Medellín, Colombia*

B.Sc. in Teaching Mathematics and Physics Jan 2004 - Oct 2009
Universidad de Antioquia *Medellín, Colombia*